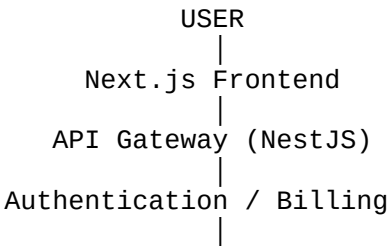


High-Level Architecture



Research Orchestrator
(Brain of the Entire Platform)

Task Planner
Workflow Engine
Agent Manager
Memory Manager
Knowledge Graph
Citation Engine
Fact Checker
Contradiction Engine
Confidence Engine
Output Generator

Autonomous Agent Layer

Market Agent
Competitor Agent
Funding Agent
Customer Agent
Technology Agent
Patent Agent
Academic Agent
News Agent
Financial Agent
Product Agent
Legal Agent
Trend Agent

Retrieval Layer

Google Search
News APIs
Academic Papers
Reddit
GitHub
Company Websites
Crunchbase
SEC Filings
Patent Databases
Books
Blogs
Internal Documents

Evidence Layer

Source Extraction
Citation Verification
Claim Extraction
Deduplication
Evidence Linking

Knowledge Layer

Knowledge Graph
Vector Database
Research Memory
Entity Database
Relationship Graph

LLM Layer

GPT
Claude
Gemini
Open Weight Models

Model Router

Storage Layer

Postgres
S3
Redis
Neo4j
Vector DB

Infrastructure

Docker
Kubernetes
Monitoring
Logging
Tracing
Queue
Caching

Complete System Modules

A mistake many founders make is thinking only about visible features.

A research operating system consists of around **30 major systems**.

1. Authentication Module

- Login
 - Signup
 - OAuth
 - Organizations
 - Teams
 - RBAC
 - API Keys
 - Sessions
 - User Preferences
-

2. Workspace Module

Projects

Folders

Tags

Archives

Templates

Shared Projects

Permissions

Version History

Activity Timeline

3. Research Request Module

Research Objective

Research Type

Depth

Budget

Deadline

Language

Country

Audience

Output Type

Custom Instructions

Priority

4. Research Planning Engine

This is the first moat.

Input:

Analyze AI coding market

Planner creates

Questions

Sub Questions

Required Sources

Required Agents

Expected Outputs

Dependencies

Execution Graph

Not a list.

A DAG (Directed Acyclic Graph).

5. Workflow Engine

Responsible for

Execution Graph

Retries

Scheduling

Parallel Execution

Dependencies

Timeouts

Checkpointing

Resume

Cancellation

Progress

6. Agent Manager

Starts agents

Stops agents

Retries

Health monitoring

Resource allocation

Agent communication

State synchronization

Agent registry

Agent capabilities

7. Agent Marketplace

Every agent is independent.

Example

Market Agent

Funding Agent

Patent Agent

GitHub Agent

Academic Agent

Hiring Agent

Pricing Agent

Security Agent

Compliance Agent

Customer Review Agent

Each agent owns

Tools

Prompt
Memory
Planner
Validator

8. Retrieval Layer

Google
Bing
News
Arxiv
PubMed
Reddit
GitHub
Crunchbase
SEC
Wikipedia
Blogs
RSS
YouTube
Podcasts
Whitepapers
Books
Internal Docs
PDFs
CSV
Excel

9. Source Normalization

Every source becomes
Document

Metadata

Author

Publisher

Timestamp

Type

Language

URL

Content

License

Reliability

10. Evidence Engine

This is the moat.

Every paragraph becomes

Claims

Each claim links to

Evidence

Each evidence links to

Sources

Example

Claim

Cursor raised \$900M.

↓

Evidence A

Evidence B

Evidence C

↓

Confidence

11. Citation Engine

Supports

APA

MLA

Chicago

IEEE

Bluebook

Inline

Footnotes

Endnotes

Automatic formatting

12. Claim Extraction Engine

Every document

↓

Split

↓

Claims

↓

Evidence

↓

Facts

↓

Entities

13. Fact Verification Engine

Cross verifies

Claims

Across

Multiple sources

Checks

Consistency

Freshness

Authority

Duplicates

14. Contradiction Engine

Finds

Conflicts

Explains

Reasons

Calculates

Confidence

Stores

Alternative viewpoints

15. Confidence Engine

Confidence score based on

Source authority

Agreement

Recency

Evidence count

Primary sources

Citation quality

16. Knowledge Graph

Entities

Companies

People
Products
Markets
Technologies
Investors
Countries
Regulations
Relationships
Everything connected.

17. Research Memory

Stores
Past projects
Past findings
Summaries
Evidence
Preferences
User decisions
Versions

18. Vector Search

Semantic Search
Hybrid Search
RAG
Embeddings
Chunking
Metadata Filtering

19. Entity Resolution

Recognizes

OpenAI

Open AI

OpenAI Inc .

Same company.

20. Report Generator

Produces

Executive Summary

Deep Report

Investment Memo

Market Report

Technical Review

SWOT

PESTLE

Porter's Five Forces

Competitive Matrix

Landscape Analysis

21. Presentation Generator

Auto PPT

Speaker Notes

Charts

Diagrams

Icons

References

22. Visualization Engine

Market Maps

Knowledge Graphs

Funding Timeline

Competitor Matrix

Heatmaps

Trend Charts

Network Graphs

Mind Maps

23. Cost Optimizer

Critical.

Chooses

GPT

Claude

Gemini

Open Source

Based on

Price

Latency

Quality

Context Window

24. Prompt Management

Prompt Library

Versioning

A/B Testing

Evaluation

Prompt Templates

25. Model Router

Not every request goes to GPT.

Some go

Claude

Gemini

Llama

Qwen

Mixtral

Based on task.

26. Evaluation Engine

Measures

Hallucination

Citation Accuracy

Coverage

Completeness

Answer Quality

Latency

Cost

27. Observability

Logs

Agent traces

Execution graph

Prompt logs

LLM outputs

Errors

Latency

Costs

28. Billing

Subscriptions

Usage

Credits

Invoices

Organizations

Rate limits

29. API Platform

Public API

Webhooks

SDK

MCP Server

CLI

30. Admin Dashboard

Users

Projects

Costs

Failures

Models

Sources

Analytics

Feature Flags

Health

Core Data Model

Organization



Workspace



Project



Research



Execution



Tasks



Agents



Sources



Documents



Claims



Evidence



Entities



Knowledge Graph



Reports

Recommended Tech Stack

Frontend

- Next.js (App Router)
- TypeScript
- Tailwind CSS
- shadcn/ui
- TanStack Query

- Zustand

Backend

- NestJS (modular monolith initially)
- Prisma ORM
- PostgreSQL
- BullMQ (job orchestration)
- Redis

AI Layer

- OpenAI API
- Anthropic API
- Google Gemini API
- LiteLLM (unified model gateway)
- LangGraph (stateful agent workflows)
- Model Context Protocol (MCP) for tool integration

Storage

- PostgreSQL (system of record)
- Neo4j (knowledge graph)
- Vector database (e.g., pgvector initially; migrate to dedicated vector DB if needed)
- S3-compatible object storage (documents, exports)
- Redis (cache and queues)

Search & Retrieval

- Tavily or Exa (web search)
- Firecrawl (website extraction)
- Jina AI Reader (content extraction)
- Unstructured (document parsing)
- OCR for scanned PDFs

Infrastructure

- Docker
- Kubernetes

- NGINX/Traefik
 - GitHub Actions
 - OpenTelemetry
 - Prometheus
 - Grafana
 - Loki
-

Development Phases

Phase 0 — Platform Foundation

- Authentication
- Organizations
- Workspaces
- Projects
- Storage
- API Gateway
- Background jobs
- LLM abstraction
- Observability

Phase 1 — MVP (8–12 weeks)

- Research objective intake
- AI research planning
- Parallel research agents
- Web/document retrieval
- Source normalization
- Claim extraction
- Citation verification
- Contradiction detection
- Confidence scoring
- Executive summary
- Deep report generation

- Export to PDF/Markdown/DOCX

Phase 2 — Research Workspace

- Persistent projects
- Research memory
- Semantic search
- Entity extraction
- Knowledge graph
- Visual dashboards
- Collaboration and comments

Phase 3 — Professional Platform

- Custom agent marketplace
- Presentation generation
- Enterprise RBAC
- Public API
- MCP server
- Integrations (Slack, Notion, Google Drive, GitHub)

Phase 4 — Intelligence Platform

- Cross-project reasoning
 - Organization-wide knowledge graph
 - Continuous monitoring of topics
 - Automated research updates
 - Fine-tuned evaluation pipelines
 - Proprietary research dataset accumulation
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